

Lab - Working with CSV Files

Objectives

Part 1: Opening a CSV File

Part 2: Creating a CSV File

Part 3: Fixing Common Errors in Importing CSV Files

Background / Scenario

CSVs are not the same as traditional spreadsheets because they do not have the saved cell, column, or row values, nor do they contain formulas. CSV files can contain numbers and letters only, even though in Microsoft Excel or Google Sheets the CSV file may look like a traditional spreadsheet.

CSV files are plain text files, which makes them easier for a developer to create, and easier for you to import into a spreadsheet or storage database regardless of software.

CSV files are easy to organize and manipulate and are primarily used for importing and exporting information to and from the source database. They are also easy to convert into traditional file formats like .xls/.xlsx, .txt, or .html.

Usually, the first row in a CSV file contains the table column labels. Each of the subsequent lines represents a row of the table. Commas separate each cell in the row, which is where the name “Comma Separated Values” comes from.

Required Resources

Mobile device or PC/laptop with a browser, Microsoft Excel, and internet access.

Note: The precise steps to format and manipulate data in Excel can vary between platforms and versions. The instructions in this lab are based on the free version of Excel available from Office.com and may have to be modified to match the user’s platform, software, or version to achieve the results shown in this lab.

Instructions

Part 1: Opening a CSV File

Step 1: Open the Bike Sales_Lab.csv file in a text editor.

- a. Open the downloaded **Bike Sales_Lab.csv** in Notepad or another text editor. Right click the file and select **Open With > Notepad**.

Alternatively, launch Notepad and select **File > Open** then navigate to the directory containing the Bike Sales_Lab.csv file. Change Notepad to look for **All Files**, then select the Bike Sales_Lab.csv file and click **Open**.

- b. Review the data in the text editor to answer the following questions:

How many columns are in the file? What are the column headings?

Answer Area

6 columns; Date, Age_Group, Product_Category, Order_Quantity, Unit_Price, Revenue

What is the delimiter used to separate the columns?

Answer Area

Comma separated

- c. Close the text editor when done.

Step 2: Open the Bike Sales_Lab.csv file in Excel.

Open the **Bike Sales_Lab.csv** in Excel

How many columns are displayed in Excel? Are these the same columns that you noted in the previous step when opening the file in a text editor?

Answer Area

Same with 6 columns

Part 2: Creating a CSV File

Many applications or websites require data uploaded as a CSV. The free version of Excel from Office.com does not support saving as a .csv file; however, the full version of Excel and the free OpenOffice spreadsheets software, called Calc, both do.

Please Note: The steps that follow are true for either the full version of Excel or OpenOffice Calc.

Step 1: Launch a paid version of Excel or the free OpenOffice Calc.

Step 2: Enter data into a blank worksheet.

Enter the following data into a worksheet as shown.

	A	B	C	D	E
1	Date	Product_Category	Order_Quantity	Unit_Price	Revenue
2	12/1/2021	Bikes	4	\$2,295.00	\$8,996.00
3	12/1/2021	Bikes	1	\$3,578.00	\$3,435.00
4	12/1/2021	Bikes	1	\$2,443.00	\$2,272.00
5	12/1/2021	Bikes	1	\$2,320.00	\$2,227.00

Step 3: Save the worksheet as a CSV file.

- To save your worksheet as a CSV file, select **File > Save As**. Choose a destination directory, name the file "**MyCSV**", and save it as a **CSV (Comma delimited)(*.csv)** if you're using Excel, or as **Text CSV (.csv)(*.csv)** if using OpenOffice.
(1) If using OpenOffice, you will be prompted with a pop-up window about file formatting; click the **Keep Current Format** button. Next, in the **Export Text File** window, keep the defaults and click **OK**.
- After saving the file, navigate to the destination directory, and open it in Notepad or another text editor. The file should look similar to the figure below.
The file is ready to upload to the desired application.

```
Date,Product_Category,Order_Quantity, Unit_Price , Revenue
12/1/2021,Bikes,4,"$2,295.00 ","$8,996.00 "
12/1/2021,Bikes,1,"$3,578.00 ","$3,435.00 "
12/1/2021,Bikes,1,"$2,443.00 ","$2,272.00 "
12/1/2021,Bikes,1,"$2,320.00 ","$2,227.00 "
```

Part 3: Fixing Common Errors in Importing CSV Files

If you work with importing CSV files, you will eventually run into input errors. Data can be messy, and imports do not always work flawlessly.

When importing CSV files, there are a few common issues that often occur, and are detailed below.

Incorrect File Type Error

If you attempt to import a CSV file without the .csv extension, you will get an error like the one below. This could happen if you created your file in a text editor and forgot to save it as a ".CSV" file type (instead, it may be a ".TXT" file). Or, perhaps you used Excel and forgot to save as CSV, but just left it as the default ".xlsx".

You attempted to upload file type that is not allowed. Please upload files in .csv format only.

If you received the above error, how would you change the extension to fix the error, assuming the data contents are correct?

Answer Area

Just resave the file to a csv file

Missing Header Row Error

This is also a very common error if the expected header titles are not in row 1. The message received will be **invalid header, no header found**, or something like this:

This file contains the following formatting errors:

- The specified CSV file is invalid. The first 6 columns must be ID, Title, Year Released, Genre, Director, and Gross.

Please fix the errors and try to upload again.

To fix this error, you need to open the .csv file and add the expected headers into row 1.

How would you add the missing header row in Excel?

Answer Area

Open the CSV file in Excel then insert a blank row above row 1 and then enter the column headers in the new row 1. Save the file and export it as a CSV file.

How would you add the missing header row in a text editor?

Answer Area

Just enter a new line and make sure you use the correct delimiter like a comma that the file is using

Missing Information in Columns and/or Rows

Another common error involves missing data in the columns or rows of the data set. Typically, if you receive an error to the effect of "empty column found" (or similar) it is because you have a column header with no data in that column or data missing from cells in a column. In other words, your first line has more data, and more delimiters, than one or more of the subsequent lines. Below is an example of an error message:

This file contains the following formatting errors:

- Line 2: No Title

Please fix the errors and try to upload again.

This error can be fixed after adding the missing data, where appropriate, in the empty cells in Excel.

How would you locate the empty cells in Excel? Perform an internet search for **how to find blank cells in Excel** if necessary.

Answer Area

One way is to open the CSV file in Excel. Click Home > Find & Select > Go To Special. Select Blanks and click OK to highlight all the found blank cells. Enter the missing data, where appropriate, and save the file again as a CSV file.

Challenge Activity

So far, most of the data sets in the activities have been relatively small. How would you locate all the blank entries in a large worksheet where you cannot easily see if you have identified all the blank cells?

What Excel functions can help you ensure that all the blank cells are located? Perform an internet search as needed.

Answer Area

To locate all blank entries in a large worksheet, the most efficient method is using the Go To Special feature. Select your data range, press Ctrl + G (or F5), click Special, select Blanks, and click OK; this instantly highlights every empty cell within the selection. Or you can use a =COUNTBLANK(range).